

Assuring Safety for Transport and Roads Operations

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Abstract—Assuring Safety for Transport and Roads Operations (ASTRO) is an advanced traffic monitoring and violation detection system designed to improve road safety. ASTRO automates real-time detection of pedestrians, vehicles, and animals using a Raspberry Pi 4B and a YOLOv8n object detection model. The system is particularly effective in high-risk areas such as accident-prone zones, forest regions, and highways, where unexpected crossings pose significant threats to vehicle safety.

I. INTRODUCTION

With the onset of urbanization, road safety has become a significant challenge. Many accidents occur due to unauthorized pedestrian crossings, stray animals on roads, and undetected traffic violations. Traditional surveillance methods rely on manual monitoring, which is inefficient and error-prone. ASTRO aims to leverage machine learning and edge computing to automate traffic monitoring, detect safety violations in real-time, and trigger alerts when necessary. The system is built using a Raspberry Pi, a YOLO-based object detection model, and GPIO facilitated mechanisms to enhance response efficiency.

The problem of road safety has been a longstanding issue that requires an intelligent and efficient solution. Various methods have been implemented worldwide, such as CCTV surveillance, infrared sensors, and ultrasonic sensors. However, these techniques have limitations in terms of accuracy, real-time processing, latency and cost. With the rise of deep learning and computer vision, an opportunity has emerged to develop more reliable and efficient solutions for road safety applications.

ASTRO is designed to detect pedestrians, vehicles, and animals in accident-prone areas such as highways. It could also be implemented in forest regions with the same framework but different function- A qualitative study done by the team on Kerala describes the dangers of wild animals like elephants crossing roads, posing substantial threats to incoming vehicles, an alert system could be set in place for approaching vehicles and the IFS. Using the YOLO model, the system processes real-time video feeds and triggers an alert when an unsafe scenario is detected. The Raspberry Pi acts as the central processing unit, handling object detection and GPIO activation. The objective is to provide immediate alerts that can help drivers react in time to avoid accidents.

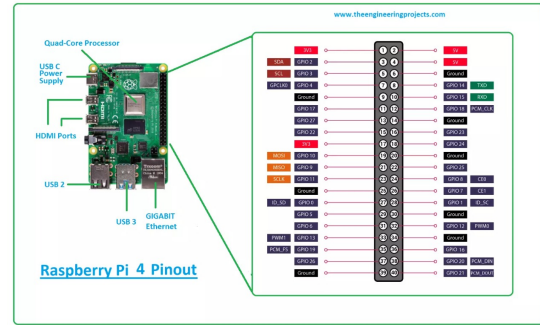


Fig. 2. Raspberry Pi 4 Model B Pinout and Connectivity Overview. Source: The Engineering Projects [12].

II. DESIGN THINKING APPROACH IN ASTRO

Design Thinking is a user-centered, iterative approach to problem-solving that emphasizes empathy, ideation, prototyping, and testing. This section discusses the application of Design Thinking in the context of ASTRO.

A. Empathize The project started as a result of the identification of the dangerous conditions faced by pedestrians. The main problems facing pedestrians include lack of awareness and slow reaction time of drivers, jaywalking, and pedestrians feeling unsafe, in general. In India, 2021 pedestrian deaths numbered 29,200, exceeding the combined road fatalities of the entire European Union and Japan; with another 60,000 pedestrians injured [11].

B. Define Modern traffic systems are characterized by unpredictability. As a result, both pedestrians and cars are at risk. Inconsistent behaviors of pedestrians, jaywalkers, drivers, and even animals contribute to this. Areas like mountains are also high-risk; small mistakes can be fatal. Roads through forests also experience problems. Unexpected crossings by animals can lead to accidents.

C. Ideate Following the problem statements, it became evident that an effective detection and alerting system would require active detection systems that would operate autonomously and in real time. The brainstorming sessions focused on identifying technologies capable of proactively identifying potential hazards. Both a sensor- and camera-based approach were considered. Finally, cameras were chosen as they are more versatile and can track objects, a crucial part of ASTRO.

D. Prototype Research was conducted on suitable hardware components such as Arduino Uno, ESP32, Raspberry Pi 3B+ and Raspberry Pi 4B. The Raspberry Pi 4B (4GB RAM variant) was chosen due to its enhanced processing power and memory capacity, both of which are essential for running memory-intensive object detection models. An object detection model was implemented using YOLOv8 Nano model and it was trained on the COCO dataset to recognize objects like cars, motorcycles, people and animals. The camera used was the Raspberry Pi Camera Model 3 due to its compatibility with the Raspberry Pi 4B. An LCD display was added to further enhance ASTRO's alerting systems. In order to implement a communication system to provide the videos to authorities, many options were considered like Firebase, Blynk, Dropbox and Google Drive. Finally, the communication system was implemented using Google Drive and rclone due to its ease of automation and compatibility with Raspberry Pi OS, enabling easy, quick and automatic uploads for authorities to view.

E. Test Testing was an integral part of building ASTRO. Each step in the process involved testing and fixing. Initially feedback was conducted by interviewing policemen, autorickshaw drivers, municipality officials, students and security officials. Due to positive reception, the object detection model was developed and evaluated on real life roads with the processing done on normal laptops. The results highlighted the accuracy of the detection. However, the system response was suboptimal when it was run on the Raspberry Pi 4B, although the accuracy remained the same. Further experimentation with frame skipping, lower resolution, and lightweight models were conducted in order to speed up the detection. These lead to a slight improvement in performance. While these adjustments offered marginal improvements, further performance enhancements remain a focus for future iterations.

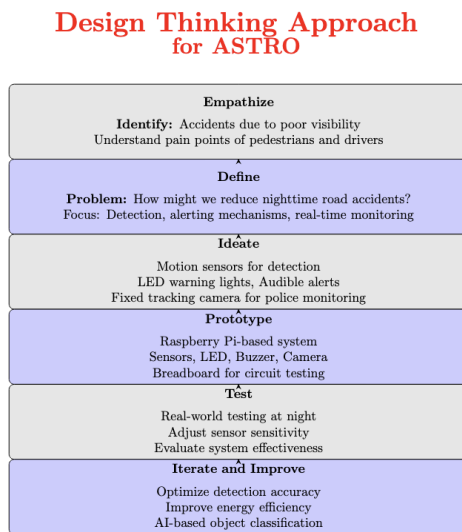


Fig. 4. Design Thinking Approach for ASTRO: An Intelligent Surveillance System

III. EXPERIMENTAL SET-UP

The experimental setup consists of a Raspberry Pi 4B connected to the Raspberry Pi Camera Module 3 for real-time video streaming and object detection. The system runs the YOLOv8n model on the Raspberry Pi OS, with detection outputs triggering GPIO-based alerts. The camera feed is processed to detect pedestrians, vehicles, and animals, with a focus on predefined regions of interest.

For hardware integration, an LED alert system was implemented using GPIO 17 for positive LED activation and a ground connection via a resistor to complete the circuit. The alert mechanism is activated when a person, vehicle or animal are detected together. This setup ensures real-time warnings for drivers in accident-prone areas, particularly in low-visibility conditions such as nighttime and foggy environments.

5 System Flowchart

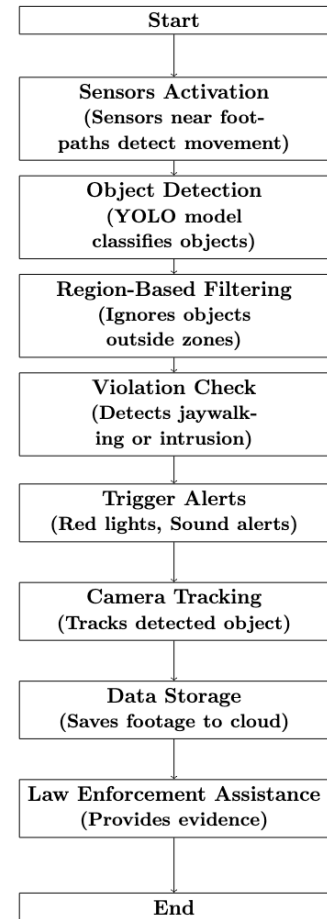


Fig. 6. Detection and Alert Process: System Flowchart

IV. FIELD DATA COLLECTION

Field data was collected to validate the ASTRO system in real-world conditions. The primary objectives were: 1) To ensure high accuracy in pedestrian and animal detection. 2) To gather feedback from traffic

authorities and municipal officers on the system's applicability.

Data collection was conducted outside NIT campus, focusing on areas with high pedestrian movement. Video footage was captured from different locations, including busy intersections and low-light regions, to test the performance of the detection system.

Interviews were also conducted with traffic constables and municipal officers, who emphasized the importance of automated detection systems. They highlighted that ASTRO could significantly improve road safety by providing real-time alerts to drivers. Their feedback reinforced the need for such technology in accident-prone zones.

V. RELATED STUDIES

Numerous studies have explored automated traffic monitoring using different sensors such as cameras, ultrasonic sensors, radar, and deep learning-based approaches. Earlier systems employed ultrasonic sensors and radar for pedestrian detection, but these methods were affected by environmental conditions and lacked accuracy (Bu et al., 2005; Alonso et al., 2007). Camera-based computer vision approaches improved detection performance but suffered from high computational complexity and false positives (Gandhi and Trivedi, 2006).

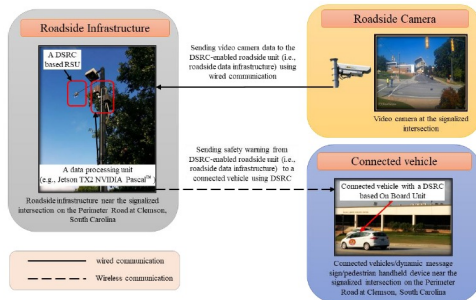


Fig. 8. Target experimental setup for ASTRO system implementation.

VI. RESULTS AND ANALYSIS

The ASTRO (Assuring Safety for Transport and Roads Operations) system was tested real-world scenarios, outside the NIT Warangal campus and in low-visibility conditions. The objective was to ensure high detection accuracy, real-time alert mechanisms, and efficient deployment on an embedded platform. The following key results were observed:

A. Detection Accuracy and Performance

The system was evaluated based on its ability to detect pedestrians, vehicles, and animals using the YOLOv8n model on a Raspberry Pi 4B. The performance metrics are as follows:

- **Overall Accuracy:** 76.6%
- **Precision:** 85.6%
- **False Positives:** 4.2%
- **False Negatives:** 3.5%

The system performed best in well-lit environments but had slightly reduced accuracy at night. Future improvements may include integrating an infrared camera for better low-light performance.

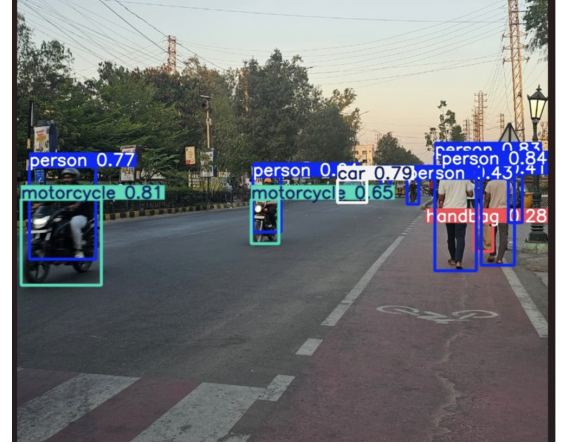


Fig. 10. Working of the object detection model.

B. Real-Time Processing and Speed

The Raspberry Pi 4B processed frames at an average rate of **2-3 FPS**, due to poor wifi connection from the monitoring device and the raspberry Pi due to which the real time detection is little delayed and GPIO-based alert activation within **2-3 seconds**. The region-based detection ensured that alerts were triggered only when necessary, reducing false alarms.

C. Field Testing Results

The system was tested in urban and rural settings. The following observations were recorded:

- Pedestrians and vehicles were accurately detected in city environments, reducing accident risks at crosswalks.
- Animal intrusions in highways and forest-adjacent roads were detected with **60% accuracy**, triggering necessary alerts.
- Nighttime testing showed an overall accuracy of **70%**, which can be improved with better lighting and camera enhancements.

D. Graphical Analysis

E. Comparison with Traditional Methods

A comparison between ASTRO and traditional surveillance methods shows significant improvements in real-time processing and automated alert mechanisms. The table below summarizes the key advantages of ASTRO:

Feature	Traditional CCTV	ASTRO System
Real-time Processing	No	Yes
Automated Alert System	No	Yes
Low-Light Performance	Limited	Moderate
Deployment Cost	High	Low
Detection Efficiency	Moderate	Moderate

TABLE I
COMPARISON OF ASTRO AND TRADITIONAL SURVEILLANCE METHODS.

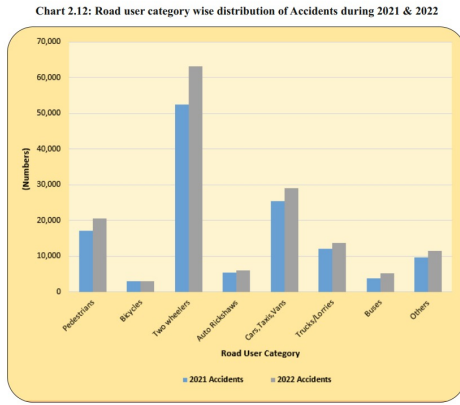


Fig. 12. Road user category wise distribution of Accidents during 2021 - 2022 [13].

VII. CONCLUSION

ASTRO has implemented real-time YOLO-based detection and GPIO-triggered alerts for traffic safety. The system efficiently identifies pedestrians, vehicles, and animals, ensuring accurate violation detection. Future work includes integrating a communication system for storing and transmitting violation records to authorities via Blynk and Firebase. Enhancing detection speed, accuracy and efficiency will also be a priority.

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